



The latest Radeon HD 4890 and GeForce GTX 285 GPUs are cooler than their predecessors and don't require additional cooling



Copper makes for the ideal component when designing heatsinks. Unfortunately it's prone to corrosion especially in contact with moisture

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depends on the assembly of the unit. Most CPU heatsinks are built to accommodate either 120 mm or 92 mm fans. Obviously, the smaller the fan the smaller the heatsink will be. These heatsinks are also called tower heatsinks owing to their height, which is much more than normal stock heatsink solutions.

When shopping for a CPU cooler one also needs to take into consideration the depth of the cabinet. Since these coolers are high, some of them might interfere with the side panel fitting – please make a note of this when shopping for one. Also most of these coolers use copper for the base, (the part that contacts the heat spreader on the CPU). This is because copper is a superior conductor of heat compared to aluminium or steel. In case you are living in a coastal region, ensure that the cooler you choose has some sort of plating (usually nickel) on the heatpipes and fins since copper corrodes with moisture. Most after-market CPU coolers come with a fan, although a few heatsinks are bundled without fans. Enthusiasts prefer these since the choice of fan is flexible while for most users the fan is another expense. Remember that the larger the fan the better because larger fans have a higher cfm rating i.e. they move more air at the same rpm, therefore cool more efficiently and with less noise.

When looking at a heatpipe-based cooler, remember to check and see how the heatpipes are attached to the base and the fins. If things are glued together it's not a good product, since any sort of glue or resin is not a good conductor of heat and will impede cooling. Ideally, the heatpipes should be soldered. Liquid coolers are also available although these are generally a bit more complicated to install, costlier and require more space inside the cabinet. I feel liquid coolers are unnecessary for most users except the hardcore overclocker who wants to benchmark components at much-higher-than-default clock speeds.

CoolerMaster is one of the brands offering a wide range of CPU coolers. Some are basic

models with slightly better specifications than the stock coolers, while others are full blown coolers with large fans, heatpipes and such. Thermalright is another brand that caters to the high-end user with some high performance CPU coolers. The default LGA Intel 775 CPU cooler will set you back by around Rs. 500, while the AMD Socket AM2+ cooler retails for around Rs. 400 – there is another one for Rs. 600, this one has two embedded heatpipes. CoolerMaster's Hyper TX 2 Super is the first step-up from the regular heatsinks and comes with a 92 mm fan and three copper heatpipes. It's ordinary looking, but built well and ships with brackets and adapters for both Intel LGA 775 and AMD sockets. Priced at Rs. 1,600, this cooler is effective, certainly better than the stock coolers, but it won't be very effective at cooling overclocked quad cores. CoolerMasters Hyper 212 Super is the next option.

Heat, dust and humidity are together your PC's worst enemies

Priced at Rs. 2,500 this cooler comes with one 120 mm fan but can use another in a push-pull configuration. It has four heatpipes and this coupled with the larger surface area of the fins brings about greater efficiency in dissipating heat, especially when using two fans.

There are two other newer entrants from CoolerMaster, the Hyper N520 and the Hyper N620. The former is priced at Rs. 2,850 and features two 92mm fans in a push-pull setup. The latter uses a beefier heatsink and has two 120mm fans. Obviously the bigger the better, but for some people fitting such a large cooler may be a problem, and in such cases I highly recommend the Hyper N520 – I've tested this and found it to be very efficient, cooling even a QX9650 overclocked to 4 GHz to just 50 degrees on load. The Hyper N620 should be more efficient and is priced at Rs. 3,260, but weighs a hefty 847 grams, enough to damage your motherboard if you lug your cabinet around with the cooler attached. The Hyper

N520 is also heavy at 688 grams, certainly heavier than most 92mm fan-based coolers.

Thermalrights Ultima 90i is the one I use on my Q9650, and this cooler is priced at Rs. 2,400. It's superbly built with six heatpipes and everything is nickel plated to avoid corrosion. Unfortunately, this excellent heatsink doesn't come with a fan – a minor irritation compared to its superb cooling. A good 92mm fan from the likes of NMB will cost you Rs. 600. For Rs. 3,000 you will have a cooler that cools a Q9650 running at 4.2 GHz to 53 degrees, and at stock speeds, (3.0 GHz), my CPU never crosses 48 degrees.

The Ultra 120 Extreme is a heatsink based around a 120mm fan and is the pinnacle of the Thermalright range at Rs. 3,400 and represents one of the best attempts at a top performing air-based CPU cooler, I've tested it, and can safely say that it's the choice if you want the best overall

performer. Unfortunately, a good 120mm fan will often cost up to Rs. 850, meaning you'll spend a little over Rs. 4,000 for this solution, placing it embarrassingly close to a few of the liquid cooling solutions around. CoolerMasters Aquagate S1 is the only liquid cooler I could find readily in stock with Lamington Road dealers; CM is the only brand with liquid coolers available all over India. The Aquagate S1 has a 120 mm fan and adapters for AMD and Intel CPUs. The block is pure copper and the S1 makes sense for crazy overclockers. There are also a few vendors who import liquid cooling kits to order. In fact, one could order a discrete pump, tubes, fan, radiator and coolant from the likes of Swiftech, which is one of the better known builders of liquid cooling equipment. However, such kits will cost at least Rs. 8,000, meaning they're for hardcore enthusiasts with cash to burn.

If you purchase enthusiast-grade memory, chances are the memory sticks will come with heatsinks. I don't advocate

slapping heatspreaders (available with many local vendors) on just about any memory. Value memory doesn't require any heatsink to dissipate heat and any heatspreader will likely trap what little heat is generated and cause more problems. OCZ has a memory cooler called the Memory Cooler XTC available for Rs. 1,200; it's got two blue-LED fans and will cool all four DIMM slots. It sits atop the RAM modules blowing air on to them. Other than this cooler there are no other active (fan based) memory coolers available, save the Corsair Dominator cooler that ships with Corsair's Dominator memory range.

GPU coolers are slightly more exotic since PCBs and GPUs themselves within a particular generation have no specified size or interface. Therefore a GPU cooler built for, say, an 8800 GT won't work on an 8800 GTX, simply because the layout of the cards differ greatly. Another point worth noting is that not all GPUs need cooling, for example the GeForce 8800 GTX GPU used to generate a lot of heat, but the much faster GTX 285 is relatively cool thanks to improvements in manufacturing technology and chip fabrication. Sadly the only decent brand available in Lamington Road was Thermalright; its HR-03 and HR-03 Plus are available for Rs. 2,300 and 2,650, respectively. Unfortunately both these solutions are pretty outdated and intended for GPUs that are largely obsolete today. I couldn't find any liquid cooling kits for GPUs. There are a few solutions available in the form of simple heatsinks and small fans that local assemblers will only be too glad to mod for your specific graphics card, but I think it's fair to say such solutions are okay if you are flogging a five year old graphics card with a dead cooler. I think it's also fair to say that today's generation of graphics cards are cool enough not to require additional cooling. Possibly the next generation of DX 11 cards will raise temperatures to an extent where manufacturers may start designing chip-specific solutions.